QP Code: 25901271

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION – March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) – MG4DSCFTQ200 – FOOD CHEMISTRY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks:** 50

ANSWER KEY

Part A – Short Answer Type

(Answer any 5 questions. Each question carries 2 marks)

1. Why do bread becomes rough and dry on storage? [A] / [CO2, CO3, CO5]

Bread becomes rough and dry due to **starch retrogradation**, where gelatinized starch recrystallizes over time, leading to moisture redistribution and firm texture (staling).

2. How do proteins act as emulsifying agents in food systems?[U] / [CO2, CO3, CO5]

Proteins contain both hydrophilic and hydrophobic groups, allowing them to **adsorb at the oil–water interface**, reduce interfacial tension, and stabilize emulsions.

3. Explain the concept of antioxidant synergism[U] / [CO1, CO2]

Antioxidant synergism refers to the phenomenon where **two or more antioxidants act together**, enhancing overall antioxidant effectiveness beyond individual action (e.g., ascorbic acid regenerating tocopherol).

4. Differentiate between amylose and amylopectin.[An] / [CO2, CO3]

Feature	Amylose	Amylopectin
Structure	Linear	Branched
Bonds	α-1,4	α-1,4 and α-1,6
Solubility	Less soluble	More soluble
Gel formation	Strong	Weak

5. Differentiate between oil-in-water and water-in-oil emulsions with one example.[K] / [CO2, CO3, CO5]

Type	Description	Example
Oil-in-water (O/W)	Oil dispersed in water	Milk
Water-in-oil (W/O)	Water dispersed in oil	Butter

6. Define caustic refining[K] / [CO4]

Caustic refining is the process of treating crude oils with **alkali (NaOH)** to remove free fatty acids, phospholipids, and other impurities.

7. What is milk sugar?[U] / [CO2]

Milk sugar is **lactose**, a disaccharide composed of glucose and galactose.

Part B

Short Essay Questions

(Answer any 5 questions. Each question carries 4 marks)

8. What are non-starch polysaccharides? Give its application in food.[U] / [CO2, CO3, CO5]

Non-starch polysaccharides are complex carbohydrates other than starch, such as **cellulose, hemicellulose, pectin, and gums**.**Applications:** act as dietary fiber, thickening agents, stabilizers, and gelling agents (e.g., pectin in jam).

9. Describe the effect of temperature, pH, metal ions and oxygen on enzymatic browning.[U] / [CO2]

- **Temperature:** increases enzyme activity up to optimum, then denatures enzyme
- **pH:** acidic pH inhibits enzyme activity
- **Metal ions:** Cu^{2+} activates polyphenol oxidase
- **Oxygen:** essential for oxidation reactions leading to browning

10. Explain specific gravity of fats and oils and state it's significance in purity of fats and oils.[U] / [CO2]

Specific gravity is the ratio of density of fat/oil to water.

Significance: used to assess purity, detect adulteration, and ensure quality control.

11. Write in brief on the following physicochemical property of sugars (1) Crystallisation (2) Effect of heat[U] / [CO2, CO3, CO5]

(1) Crystallisation: Sugar forms crystals depending on supersaturation and cooling; important in confectionery products.

(2) Effect of heat: Heating sugars leads to **caramelization**, producing brown color and flavor compounds.

12. Describe about the factors affecting activity of phenolase enzymes in foods.[U] / [CO2]

Factors include temperature, pH, oxygen availability, substrate concentration, and inhibitors like ascorbic acid and sulfites.

13. Define rancidity. Discuss about different types of rancidity.[K] / [CO1, CO2]

Rancidity is the deterioration of fats and oils resulting in unpleasant odor and taste. **Types:**

- Oxidative rancidity (reaction with oxygen)
- Hydrolytic rancidity (lipase activity)
- ketonic rancidity

14. Discuss the application of pectin in gel formation.[A] / [CO2, CO3, CO5] Pectin forms gels in the presence of **acid and sugar** by forming a network structure that traps water. It is widely used in jams and jellies.

Part C

Long Essay Questions

(Answer any 2 questions. Each question carries 10 marks)

15. Describe the different methods and products of starch hydrolysis and it's important applications in food.[An] / [CO2, CO3, CO5]

Methods:

- Acid hydrolysis
- Enzymatic hydrolysis (α-amylase, β-amylase, glucoamylase)

Products: Glucose, Maltose, Dextrins and High fructose corn syrup

Applications:

- Sweeteners
- Brewing industry
- Bakery products
- Thickening agents

16. Define amylase enzymes and explain its types along with its functions.[U] / [CO2]

Amylases are enzymes that hydrolyze starch into simpler sugars.

Types:

- **α-amylase:** breaks internal α-1,4 bonds
- **β-amylase:** releases maltose from ends
- **Glucoamylase:** produces glucose

Functions: Used in the digestion, brewing, baking, and starch processing industries.

17. Explain the procedure to determine peroxide value with calculations and discuss it's importance in assessing rancidity.[U] / [CO2]

Procedure:

- React fat sample with potassium iodide
- Liberated iodine titrated with sodium thiosulfate

Calculation:

Peroxide value = $(S - B) \times N \times 1000 / W$

Importance:

Indicates extent of oxidative rancidity and quality of fats.

18. Discuss the structure and modifications of cellulose in food products.[U] / [CO2, CO3, CO5]

Structure: Cellulose is a linear polymer of β-1,4 linked glucose units forming strong fibers.

Modifications: Carboxymethyl cellulose (CMC) and Microcrystalline cellulose

Applications:

Used as thickener, stabilizer, fat replacer, and dietary fiber in food products.